

ASEN 6044

Homework #3

Bijan Jourabchi

Problem 1:

- a) Given the following prior information on independent state x , measurment bias z , and measurment likelihood $P(y | x, z)$

$$p(x) = \mathcal{U}[1, 11] \quad p(z) = \mathcal{G}(a, b) = \frac{1}{b^a \Gamma(a)} z^{a-1} e^{-z/b}$$

$$P(y | x, z) = \mathcal{U}[x, (x + z)^2]$$

with $a = b = 2$. We can derive the joint distribution as $P(x, z) = P(x)P(z)$ since x and z are independent. To determine the joint posterior PDF, $P(x, z | y_1, y_2, \dots, y_n)$ for a static state (not changing), we can use bayes rule.

$$P(x, z | y_1, y_2, \dots, y_n) = \frac{P(x, z | y_1, y_2, \dots, y_{n-1})P(y_n | x, z)}{P(y_1, y_2, \dots, y_n)}$$

Thus, for the first measurment $y_1 = 91.56$ we have

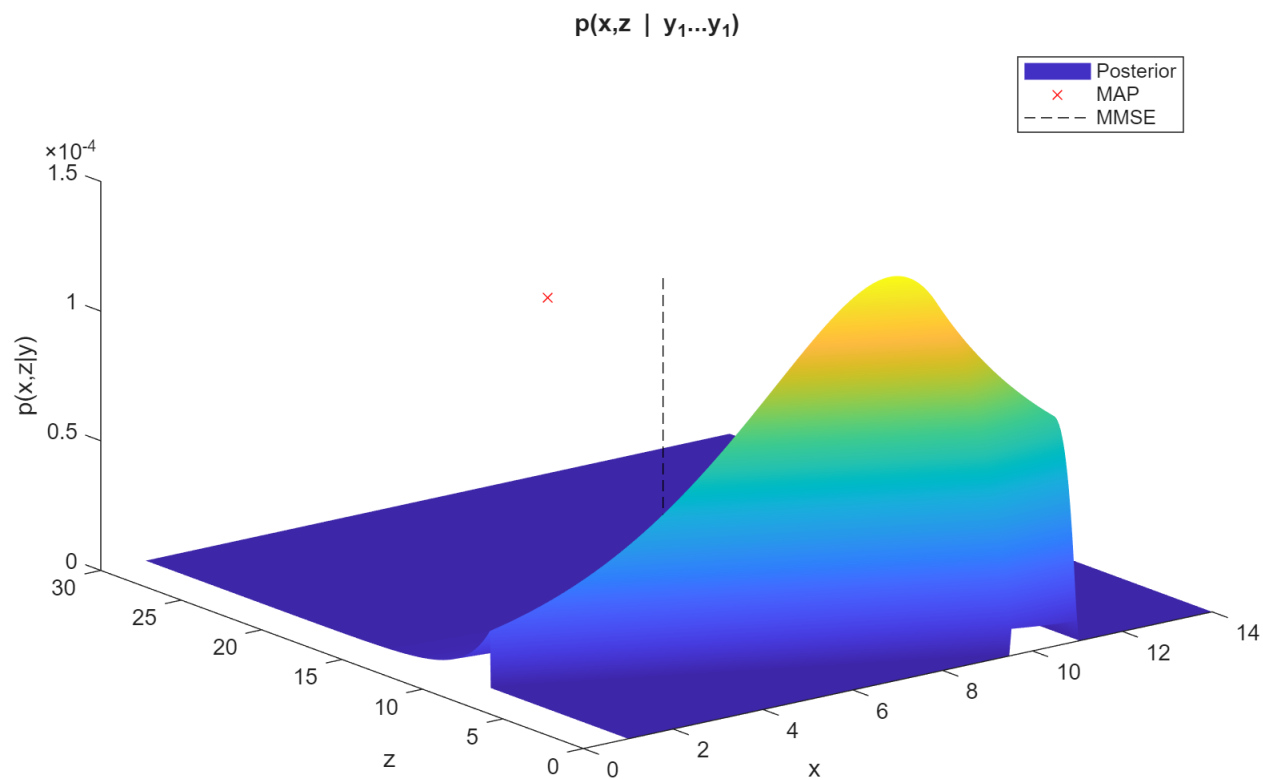
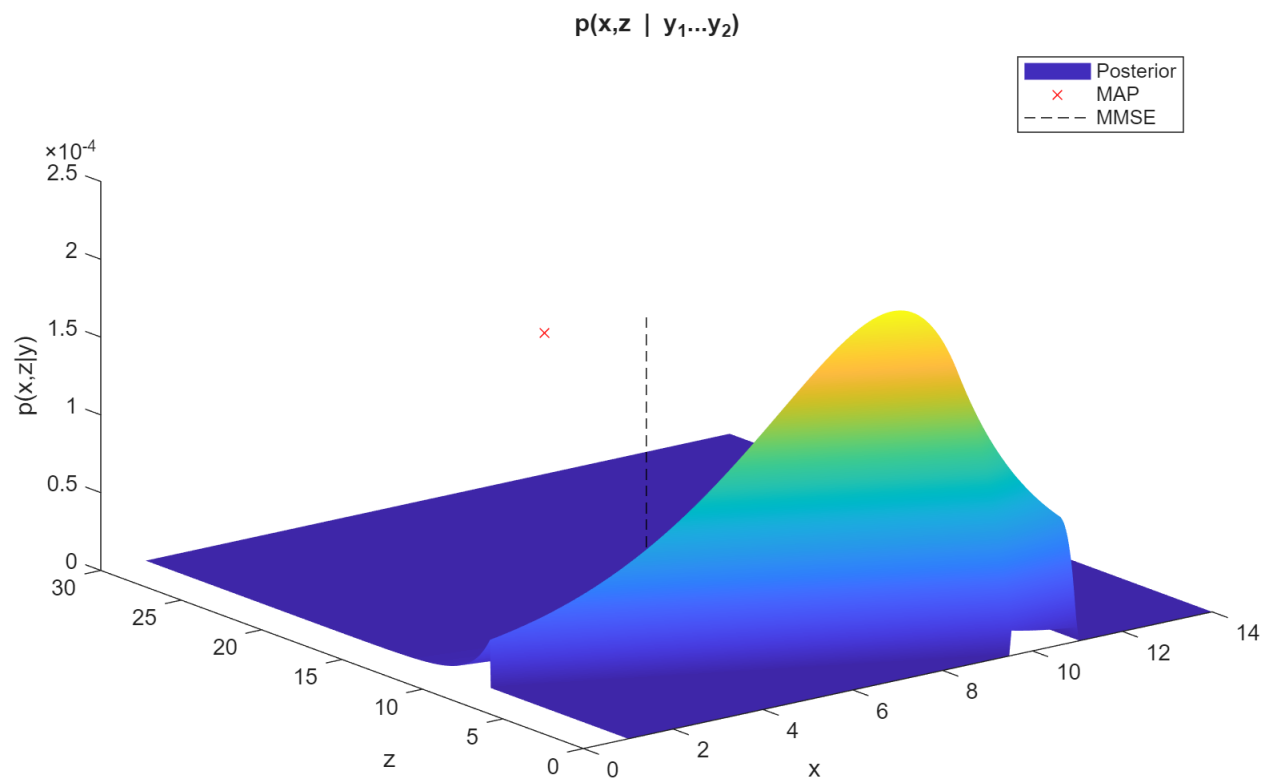
$$P(x, z | y_1) = \frac{P(x, z)P(y_1 | x, z)}{P(y_1)}$$

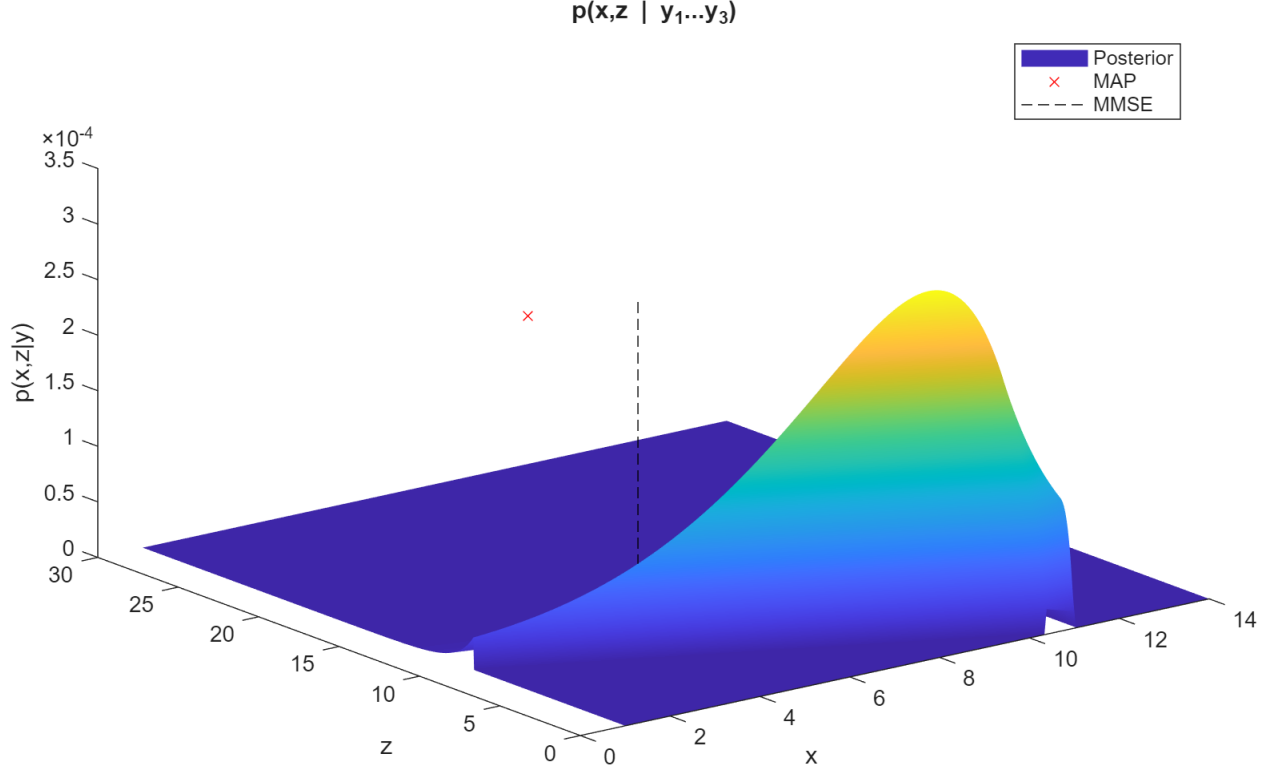
We can derive similar expressions for subsequent measurments y_2 and y_3 ,

$$P(x, z | y_1, y_2) = \frac{P(x, z | y_1)P(y_2 | x, z)}{P(y_1, y_2)}$$

$$P(x, z | y_1, y_2, y_3) = \frac{P(x, z | y_1, y_2)P(y_3 | x, z)}{P(y_1, y_2, y_3)}$$

Coding this up using the given intervals for x and z , we can plot the joint posteriors seen in figures (1),(2), and (3). We see that after each subsequent measurment, the probability density changes slightly each time. This is due to each new measurment encoding more information on what the state could be.

Figure 1: Joint Posterior $p(x, z \mid y_1)$ Figure 2: Joint Posterior $p(x, z \mid y_1, y_2)$

Figure 3: Joint Posterior $p(x, z | y_1, y_2, y_3)$

- b) Using the result from part a, we can easily determine the MMSE and MAP estimates of each joint posterior distribution. Table 1 summarizes these estimates for the three posterior distributions.

Table 1: MMSE and MAP estimates for the joint posterior distributions

Posterior	MMSE $\hat{x}$	MMSE $\hat{z}$	MAP $\hat{x}$	MAP $\hat{z}$
$p(x, z y_1)$	4.5064	7.6296	1.95	7.65
$p(x, z y_1, y_2)$	4.0683	7.4447	1.9	7.7
$p(x, z y_1, y_2, y_3)$	4.0755	7.7973	1.9	8.55

The value of the marginal data likelihoods $P(y_1), P(y_1, y_2), P(y_1, y_2, y_3)$ are the same as the normalization constants used when normalizing the joint posteriors. These values are $P(y_1) = 3.634 \times 10^{-3}$, $P(y_1, y_2) = 7.8 \times 10^{-3}$, and, $P(y_1, y_2, y_3) = 5.01 \times 10^{-3}$.

- c) Before performing Monte Carlo IS, we must derive an expression for the weights. The prior of $P(x, z | y_1, y_2, y_3)$ is just the joint posterior after receiving the second measurement, or $q(x, z; y_1, y_2, y_3) = P(x, z | y_1, y_2)$. The weights for IS are given as the ratio of the distribution we are trying to approximate, $P(x, z | y_1, y_2, y_3)$ to the proposal distribution. In this case we can simplify the expression for the importance weights.

$$w(X_s^i) = \frac{P(x_i, z_i | y_1, y_2, y_3)}{P(x_i, z_i | y_1, y_2)} \propto \frac{P(x_i, z_i | y_1, y_2)P(y_3 | x_i, z_i)}{P(x_i, z_i | y_1, y_2)} = P(y_3 | x_i, z_i)$$

Thus, the weight of a sampled point is proportional to the measurement likelihood of y_3 . Thus, the particle weight for a sampled point x_s is

$$\tilde{w}(X_s^i) = \frac{w(X_s^i)}{\sum_{i=1}^{N_s} w(X_s^i)}$$

Performing IS with 500 samples, we get the result seen in figure (4). The MMSE estimate from the IS particles is the weighted sum of the particles. Computing this, we get a MMSE estimate of $\hat{x}_{MMSE,IS} = 4.21$ and $\hat{z}_{MMSE,IS} = 7.63$. Comparing this result from the MMSE value from Table 1, we see that the two are very similar. While the MMSE estimate from IS is slightly different than that of the exact grid based solution, importance sampling allows us to only sample the important areas of the state space. The loss of accuracy is traded off for the lighter computational load.

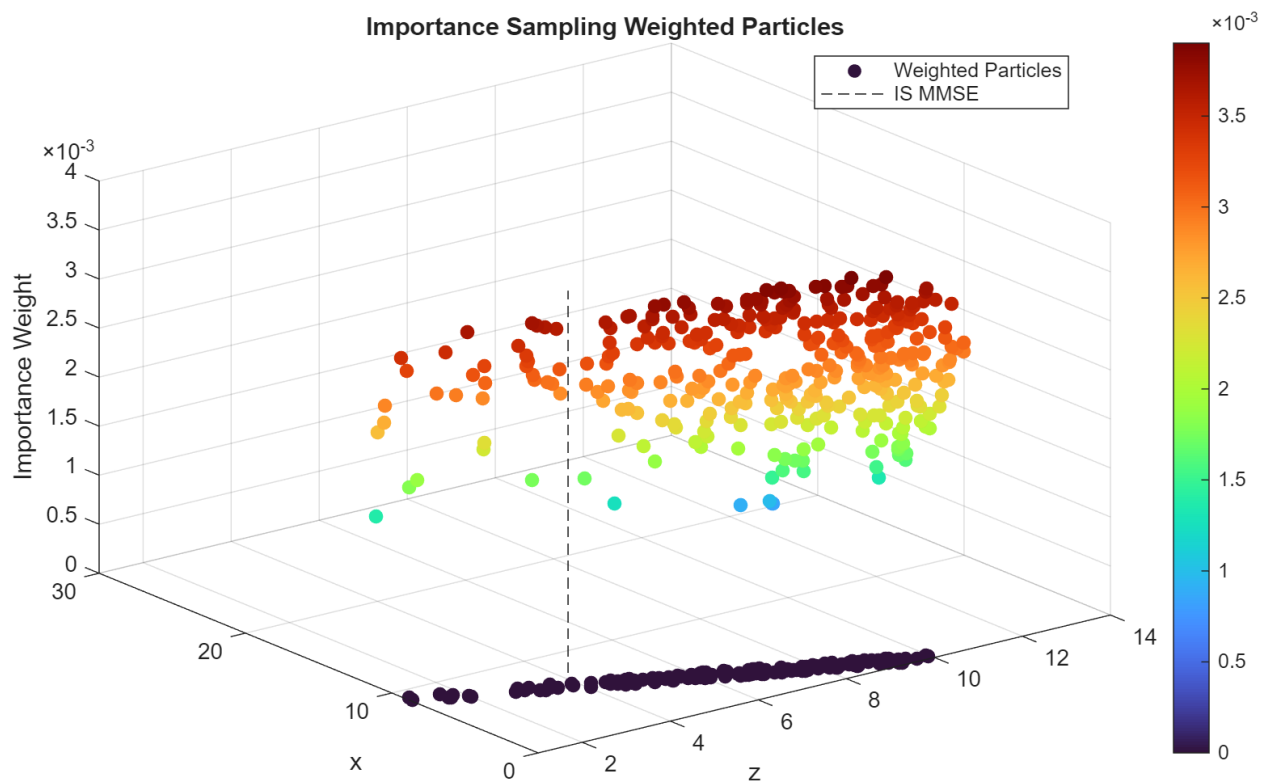


Figure 4: IS Estimate of $p(x, z \mid y_1, y_2, y_3)$

1.1 Code For Problem 1

```

1 clear; clc; close all
2
3 rng(500)
4 %% Define grid space + paramters
5
6 dx = 0.05;
7 x = 1:dx:14;
8 z = dx:dx:30;
9 a = 2;

```

```

10 b = 2;
11
12 Nx = length(x);
13 Nz = length(z);
14 %% Define prior grid dist
15
16 px = zeros(1,Nx);
17 px(x >= 1 & x <= 11) = 1/(11-1);
18
19 pz = (1/(b^a * gamma(a))) .* (z.^(a-1)) .* exp(-z./b);
20
21 P = zeros(Nx,Nz);
22
23 for i = 1:Nx
24     for j = 1:Nz
25         P(i,j) = px(i) * pz(j);
26     end
27 end
28
29 P = P / sum(P(:));
30
31 [X,Z] = meshgrid(z,x);
32
33 figure
34 surf(X,Z,P,'DisplayName','Prior')
35 xlabel('z')
36 ylabel('x')
37 zlabel('p(x,z)')
38 title('Joint Prior Distribution p(x,z)')
39 shading interp
40 legend('Location','best')
41
42 %% Posterior Bayes Update
43
44 y = [91.56, 70.43, 108.67];
45 Ny = 3;
46
47 L = likelihood(y(1),x,z);
48
49 figure
50 surf(X,Z,L,'DisplayName','Likelihood')
51 xlabel('z')
52 ylabel('x')
53 zlabel('p(y|x,z)')
54 title('Likelihood Dist p(y|x,z)')
55 shading interp
56 legend('Location','best')
57
58 for i = 1:Ny
59     Posterior = BayesUpdate(y(i),x,z,P, true,i);
60     P = Posterior;
61     MAP = getMAP(P,x,z);
62     MMSE = getMMSE(P,x,z);
63 end

```

```

64
65 %% IS
66
67 px = zeros(1,Nx);
68 px(x >= 1 & x <= 11) = 1/(11-1);
69
70 pz = (1/(b^a * gamma(a))) .* (z.^(a-1)) .* exp(-z./b);
71
72 P = zeros(Nx,Nz);
73
74 for i = 1:Nx
75     for j = 1:Nz
76         P(i,j) = px(i) * pz(j);
77     end
78 end
79
80 for i = 1:Ny-1
81     Posterior = BayesUpdate(y(i),x,z,P, false,i);
82     P = Posterior;
83     MAP = getMAP(P,x,z);
84     MMSE = getMMSE(P,x,z);
85 end
86
87 Ns = 500; % Sample Count
88
89 % Sample from prior
90 cdf_prior = cumsum(P(:));
91 random_values = rand(Ns, 1);
92 sample_indices = arrayfun(@(t) find(cdf_prior >= t, 1, 'first'),
93     random_values);
94 [row_indices, col_indices] = ind2sub(size(P), sample_indices);
95
96 xi = x(row_indices);
97 zi = z(col_indices);
98
99
100 w = zeros(1,500);
101
102 for i = 1:Ns
103     %w(i) =
104         likelihood(y(1),xi(i),zi(i))*likelihood(y(2),xi(i),zi(i))*likelihood(y(3),xi(i),zi(i));
105     w(i) = likelihood(y(3),xi(i),zi(i));
106
107 end
108
109 w = w./sum(w); % Normalize
110
111 % MMSE estimate
112 IS_MMSE = [0;0];
113 for i = 1:Ns
114     IS_MMSE = IS_MMSE + w(i)*[zi(i);xi(i)];
115 end
116
117 figure; hold on

```

```

116 scatter3(xi, zi, w, 40, w, 'filled', 'DisplayName', 'Weighted Particles')
117 plot3([IS_MMSE(1) IS_MMSE(1)], [IS_MMSE(2) IS_MMSE(2)], [0 max(w)], 'k--',
        'DisplayName', 'IS MMSE')
118 xlim([1 14])
119 ylim([0 30])
120 xlabel('z')
121 ylabel('x')
122 zlabel('Importance Weight')
123 title('Importance Sampling Weighted Particles')
124
125 colorbar
126 colormap turbo
127 grid on
128 legend('Location', 'best')
129 view(3)
130 exportgraphics(gcf, "IS.png", 'Resolution', 200);
131
132 %% Helper Functions
133
134 function [L] = likelihood(y,x,z)
135
136 Nx = length(x);
137 Nz = length(z);
138 L = zeros(Nx,Nz);
139
140 for i = 1:Nx
141     for j = 1:Nz
142         % Out of bounds
143         if y < x(i) || y > (x(i)+z(j))^2
144             L(i,j) = 0;
145             continue
146         end
147
148         L(i,j) = 1/((x(i)+z(j))^2-x(i));
149     end
150 end
151
152
153 end
154
155 function [Posterior] = BayesUpdate(y,x,z,Prior, PLOT,i)
156
157 L = likelihood(y,x,z);
158
159 Posterior = Prior.*L;
160 fprintf("Value of Marginal P(y_1:N) = %d for meas 1:%i\n",sum(Posterior(:)),i)
161 Posterior = Posterior./ (sum(Posterior(:)));
162
163
164
165 if PLOT
166     MAP = getMAP(Posterior,x,z);
167     MMSE = getMMSE(Posterior,x,z);

```

```

168     [X,Z] = meshgrid(z,x);
169     figure
170     hold on
171     surf(Z,X,Posterior,'DisplayName','Posterior')
172     scatter3(MAP(1),MAP(2),MAP(3),'rx','DisplayName','MAP')
173     plot3([MMSE(1) MMSE(1)],[MMSE(2) MMSE(2)],[0
        MAP(3)],'k--','DisplayName','MMSE')
174     xlabel('x')
175     ylabel('z')
176     zlabel('p(x,z|y)')
177     title(sprintf('p(x,z | y_1...y_%d)',i))
178     shading interp
179     legend('Location','best')
180     view(3)
181     exportgraphics(gcf, sprintf('Measurment_%d',i) + ".png",
        'Resolution', 200);
182 end
183 end
184
185 function [MAP] = getMAP(P,x,z)
186
187     [peak,idx] = max(P(:));
188     [row, col] = ind2sub(size(P), idx);
189     [X,Z] = meshgrid(z,x);
190     map_z = X(row,col);
191     map_x = Z(row,col);
192
193     MAP = [map_z;map_x;peak];
194
195 end
196
197 function [MMSE] = getMMSE(P,x,z)
198 [X,Z] = meshgrid(z,x);
199
200 z_mmse = sum(sum(P .* X));
201 x_mmse = sum(sum(P .* Z));
202
203 [~, ix] = min(abs(x - x_mmse));
204 [~, iz] = min(abs(z - z_mmse));
205
206 MMSE = [z_mmse; x_mmse; P(ix,iz)];
207
208 end

```

Problem 2:

- a) Given the state index transition probabilities, we can compile all of the transition information into a matrix $T \in \mathbb{R}^{N_x \times N_x}$ with

$$T_{ij} = P(\zeta_{k+1} = i \mid \zeta_k = j), \quad \rho_{k+1} = T\rho_k.$$

Therefore, column j corresponds to the current state $\zeta_k = j$, while row i corresponds to the propagated state $\zeta_{k+1} = i$.

$$T = \begin{bmatrix} 0.1 & 0.2 & 0 & 0 & \cdots & 0 & 0 \\ 0.9 & 0.3 & 0.2 & 0 & \cdots & 0 & 0 \\ 0 & 0.5 & 0.3 & 0.2 & \ddots & \vdots & \vdots \\ 0 & 0 & 0.5 & 0.3 & \ddots & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & 0.2 & 0 \\ 0 & 0 & \cdots & 0 & 0.5 & 0.3 & 0.9 \\ 0 & 0 & \cdots & 0 & 0 & 0.5 & 0.1 \end{bmatrix}.$$

A stationary distribution ρ_{∞}^- such that $\rho_{\infty}^- = T\rho_{\infty}^-$ will exist if the square matrix T has an eigenvalue $\lambda = 1$. In such case, the matrix eigenvalue equation becomes $T\vec{v} = \vec{v}$ where $\vec{v}$ is the eigenvector corresponding to $\lambda = 1$. By definition of a stationary distribution, the eigenvector corresponding to $\lambda = 1$ is our stationary distribution. Since T has an eigenvalue $\lambda = 1$, there exists a stationary distribution ρ_{∞}^- .

b)